

04 AI Collaboration Note

Source: docs/case-study/AI_COLLABORATION.md

AI Collaboration Note

Status: Day 5 final

Tools used and role of each

Tool	Role
Cursor (Composer / Claude)	Architecture drafts, implementation, docs, deploy scripts
LangGraph	Agent graph with interrupt for human approval
FastAPI + Pydantic	API, structured outputs, validation
Next.js 15	Approval inbox and New Run UX
Context7	Library docs (FastAPI/Vercel/HubSpot patterns)
Deepgram (and mock STT)	Speech-to-text for audio uploads
Multi-LLM registry (Gemini, OpenAI, Anthropic, Groq, Ollama Cloud, mock)	Drafting / scope verify via BYOK
HubSpot / Gmail / Jira APIs	Read deal context; execute after approve
Turso (libSQL)	Durable production database on Vercel
Agency agents (local clone)	Product Manager / UI / UX / Backend guidance for polish

Work delegated to AI

- * [x] LangGraph workflow scaffold (scope -> draft -> grade -> approve interrupt)
- * [x] Pydantic schemas for runs, scope flags, action packages
- * [x] Scope Verifier + hallucination grader logic
- * [x] FastAPI routes (auth, runs, integrations, SOWs)
- * [x] Next.js Approval Inbox / New Run / credentials UI
- * [x] Eval fixtures (`eval/test\_cases.json`) and `eval/run\_eval.py`
- * [x] Documentation drafts (architecture, runbook, case study shell)
- * [x] Deploy wiring (Vercel API + web, Turso env, HubSpot pipeline dropdowns)

How AI-generated results were verified

- * [x] Pydantic validation on structured LLM outputs
- * [x] 12-case eval suite with pass/fail (python eval/run_eval.py -> 12/12)
- * [x] Manual root-cause write-ups for >=3 failures (FAILURES.md)
- * [x] Proxy baseline timing on three transcripts (BASELINE.md)
- * [x] Regression re-run after grader / owner-confidence / Gmail retry fixes
- * [x] Live deploy smoke checks (/health, CORS, login/session, HubSpot deal list)

Results rejected or manually corrected

Item	AI suggested	Human decision	Why
Auto-send email after draft	Ship without approve	**Rejected**	Assessment + real risk require human gate
Invite-only allowlist	Keep AUTH_ALLOWLIST enforced	**Removed**	for open register anyone can
Render free API	Primary production host	**Moved API to Vercel**	Render billing wall blocked free d eploy
Local SQLite only on Vercel	"Fine for prod"	**Added Turso**	Redeploys wiped users, vault, connecti ons
Cookie `SameSite=Lax` cross-subdomain	Default session cookie	**`SameSite=None; Secure`**	Web and A PI on dif
Free-text HubSpot `dealstage`	Type stage strings	**Pipeline stage dropdowns**	HubSpot rejects inve nted IDs;
"Create sample deal" reuse first deal	Return existing if any	**Always create new**	User needed mul tiple dea
Invented cost / P95 in RESULTS	Fill placeholder \$ figures	**Left unmetered**	Only measured times a nd suite

Core decisions owned personally

1. Product is a 4-engine OS with approval, not a meeting summarizer
2. Integrations: HubSpot + Gmail + Jira (not Salesforce)
3. Scope Verifier as the differentiator against naive ChatGPT
4. Non-goals: no auto-send, no Zoom bot, English-only v1
5. Eval rubric and the 12 test cases

6. Architecture trade-offs (transcript-in, BYOK, Turso)

7. Demo narrative: 38 -> 6 minutes, 12/12 suite, honest limitations